

Safety Data Sheet (S D S)

1. Product and company identification

GHS product identifier

(Product name)

Pre-Mix powder

(JIP KW. 06. 25T3)

Supplier' s details

Name: JFE Steel Corporation

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Iron Powder Sec. Product Design & Quality Control Dept.

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Reference number of issue J5E-1

Date of revision 13th April 2015

2. Hazards identification

<Physicochemical hazards>

Flammable solids No classification

◇Classification of the mixture :

<Health hazard>

Hazard class	Classification	Health hazard statements
Skin corrosion and irritation	Category 3	Causes mild skin irritation (H316)
Serious eye damage / eye irritation	Category 2B	Causes eye irritation (H320)
Respiratory sensitization	Category 1	May cause allergy or asthma symptoms or breathing difficulties if inhaled (H334)
Carcinogenicity	Category 2	Suspected of causing cancer (state route of exposure if it is conclusively proven that no other routes of exposure cause the hazard) (H351)
Specific target organ toxicity (single exposure)	Category 1	Causes damage to organs (H370)
	Category 3	May cause respiratory irritation (H335)
Specific target organ toxicity (repeated exposure)	Category 1	Cause damage to organs (H372)

<Environmental hazards>

Hazard class	Classification	Environmental hazard statements
Hazardous to the aquatic environment, long-term hazard	Category 4	May cause long lasting harmful effects to aquatic life (H413)

◇GHS label element :

<Pictogram or Symbol>



<Signal word>

Danger, Warning

<Notes>

(Precautionary statement)

- Do not handle until all safety precautions have been read and understood. (P202)
- Do not breathe dust/fume. (P260)
- Avoid breathing dust/fume. (P261)
- Wash hands thoroughly after handling. (P264)
- Do not eat, drink or smoke when using this product. (P270)
- Use only outdoors or in a well-ventilated area. (P271)
- Contaminated work clothing should not be allowed out of the workplace. (P270)
- Wear protective gloves. (P280)
- In case of inadequate ventilation wear respiratory protection. (P284)

(First-aid measures)

- IF ON SKIN: Wash with plenty water / water with soap. (P302+P352)
- IF INHALED: Remove person to fresh air and keep comfortable for breathing. (P304+P340)
- IF IN EYES: Rinse cautiously with water for several minutes, Remove contact lenses, if present and easy to do. Continue rinsing. (P305+P351+P339)
- Call a doctor, if you feel unwell. (P312)
- If skin irritation or rash occurs: Get medical advice / attention. (P333+P313)
- IF eye irritation persists: Get medical advice / attention. (P337+P313)
- IF experiencing respiratory symptoms: Call a doctor. (P342+P311)
- Take off contaminated clothing and wash it before reuse. (P362+P364)

(Storage)

- Store in a well ventilated place (P403)
- Store locked up (P405)

(Disposal)

- Dispose of contents/container in accordance with local/regional/national/international regulation to be specified. (P501)

3. Composition/information on ingredients

◇Chemical identity: Mixture (alloy based on iron and chemically stable solid substance)

◇Main constituents and concentration range :

Constituents		Concentration range[wt%]	CAS number
Iron	[Fe]	bal	7439-89-6
Nickel	[Ni]	Less than 5.0	7440-02-0

Copper	[Cu]	Less than 2.0	7440-50-8
Molybdenum	[Mo]	Less than 2.0	7439-98-7
Zinc stearate	[C ₃₆ H ₇ OO ₄ Zn]	Less than 2.0	557-05-1
Graphite	[C]	Less than 2	7782-42-5

Note 1) Composition values will vary within the concentration ranges mentioned above depending on the steel type/standards

Note 2) Depending on the application, products may include traceable elements other than the ingredients mentioned above, for example Carbon [C], Phosphorus [P], Sulfur [S], Nitrogen [N].

4. First-aid measures

Persons using this product should consult a physician or other medical professional if an accident involving this product in injury. Specific first-aid measures are as follows:

- ◇ **IF INHALED** : Remove person to fresh air and keep comfortable for breathing.
Call a doctor/physician if you feel unwell.
- ◇ **IF ON SKIN**: Wash with plenty water / water with soap
- ◇ **IF IN EYES**: Rinse cautiously with plenty of water at least for several minutes, Remove contact lenses, if present and easy to do. Continue rinsing. If irritation persists, seek medical care.
- ◇ **IF SWALLOWED**: Rinse mouth. Do NOT induce vomiting. If necessary, after taking first-aid measures, get medical attention immediately.

5. Fire-fighting measures

- ◇ **Fire-extinguisher media** : Water spray, dry chemical, dry sand type

6. Accidental release measures

- ◇ **Personal precautions**: Use the wearing suitable protective equipment (including personal protective equipment, see Section 8 of the SDS) to prevent iron powder on skin, eyes or personal clothing.
- ◇ **Protective equipment** : See personal protection in section 8 (Exposure controls/personal protection)
- ◇ **Environmental precautions** : No release to the environment, In particular, powder is discharged into the river, to be careful not to cause harmful effects on the environment.
- ◇ **Methods and materials for containment and clean up** : Collected by the vacuum cleaner, etc., and placed in a vessel, to prevent secondary leakage .

7. Handling and storage

◇ **Handling** :

<technical measures>

See personal protection in section 8 described.

Ensuring partial/complete ventilation.

<precaution for safe handling>

When you are working to transfer powder from the flexible container to hopper, storage, container, mixer, wear suitable protective equipment.

If dust occurs, not discharged with a cover.

If strings or ropes of flexible containers are damaged, there is a possibility of bag

breakage and falling, be careful of handling. In particular, if you hang in the flexible container in forklift, to note the curing of fork nail.

In order to keep the exposure limits below the concentration in the air is performed ventilation exhaust.

◇Storage :

<Conditions for safe storage*1>

It is noticed about stacking of flexible bags, racking stack of them, a fall depending on how to place others, collapse of cargo piles, and fall for a heavy load. Keep away from water leakage or acid/alkali liquid or materials that contain them and environments of high temperature and humidity.

If necessary, use sheet/cover/packing to prevent wetting by rain water or staining.

*1 Even if not following these procedures, there is no outbreak of dangerous noxious materials. The above procedures are to prevent a drop in the quality of iron powder products.

8. Exposure controls/personal protection

Follow the exposure control/personal protection procedures shown below.

◇Concentration limit :

		ACGIH*1	
Constituents	CAS number	TLVs-TWA [mg/m ³]	
Nickel [Ni]	7440-02-0	1.5	
Copper [Cu]	7440-50-8	1*3/0.2*3	
Molybdenum [Mo]	7439-98-7	10*2 (I)/3*2 (R)	

Note) Search results of NITE HP/CHRIP (Chemical Risk Information Platform)

*1 American Conference of Governmental Industrial Hygienists

*2 (I):Inhalable fraction (R):Respirable fraction

◇Exposure controls : If dust occurs, use partial/whole ventilation.

◇Personal protection : Use suitable respiratory protective equipment/protective gloves/ safety shoes/ safety helmet. If lot of dust occurs, use a dustproof mask.
If necessary, use a dustproof apron, protective wear.
Recommended to wear safety glasses.

9. Physical and chemical properties

◇Physical state, color etc : silvery powder

◇Odor : odorless

◇Melting point : 1535°C or higher

10. Stability and reactivity

◇Stability : Stable under normal condition

- ◇**Hazardous reactions** : On contact with chemicals acid, alkali, there is a possibility that harmful gases that are generated.
- ◇**Condition to avoid** : Is mixed with air at a ratio of explosion limits, if there is ignition source, dust explosion might occur.
- ◇**Incompatible material** : Strong oxidizing agents, acids.
- ◇**Hazardous decomposition products** : Will not occur under normal condition.

11. Toxicological information

Hazard class	Classification			
	[Ni]	[Cu]	[Mo]	[ZnSt]
Acute toxicity	—	—	—	—
Skin corrosion/irritation	—	—	—	Category 3
Serious eye damage/irritation	—	—	—	Category 2
Respiratory or skin sensitization	Category 1	—	—	—
Germ cell mutagenicity	—	—	—	—
Carcinogenicity	Category 2	—	—	—
Reproductive toxicity	—	—	—	—
STOT-single exposure	Category 1	Category 3	Category 3	Category 3
STOT-repeated exposure	Category 1	Category 1	—	—
Aspiration hazard	—	—	—	—

Note 1) Search results of NITE HP/Chemical Management Field/GHS Information

Note 2) " — " in above table means out of category or unclassifiable

Note 3) Show information of Category at section 2(Hazard identification)

12. Ecological information

Hazard class	Classification			
	[Ni]	[Cu]	[Mo]	[ZnSt]
Hazardous to the aquatic environment (acute)	—	—	—	—
Hazardous to the aquatic environment (chronic)	Category 4	Category 4	—	—

Note 1) Search results of NITE HP/Chemical Management Field/GHS Information

Note 2) " — " in above table means out of category or unclassifiable

Note 3) Show information of Category at section 2(Hazard identification)

13. Disposal considerations

Dispose of contents/container/packing material adequately in accordance with local/regional/national/international regulation of waste, when discarding industrial waste such as dust etc., or containers which receive waste, or packing material with which products is packed up at shipment.

14. Transport information

Follow all regulation on the transport in your country or region.

15. Regulatory information

- ◇Industrial safety and health law
- ◇Law about the promotion of the improvement of monitoring and controlling of the discharge to environment of specified chemical substances and the management.

16. Other information

◇References:

- Hazardous Substance Data Book (Tokyo Fire Department, Research Committee)
- Metallic Alloys and Harmonization of Classification Criteria (OECD)
- Safety Data Sheet for Chemical Products (ISO 11014-1)
- Recommended values of the Japan Industrial Hygiene Society, OSHA PEL, ACGIH TLV, etc.
- Industrial Safety and Health Law
- PRTR (Pollutant Release and Transfer Registers)

The purpose of this material safety data sheet is to provide business owners that use the products with reference information to ensure chemically safe handling. This sheet is prepared based on materials and information that is available to us at the present time. Therefore, there might be hazard or harmfulness beyond our knowledge and so this sheet is not a guarantee of safety. It is appreciated that businesses owners who handle these products are requested to understand the necessities of taking appropriate safety measures in accordance with the applications and methods of usage and to use the contents as written herein.